

Answer ALL questions.

Write your answers in the space provided.

Some questions must be answered with a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 A plant contains several types of cells, tissues and organs.

(a) State what is meant by the term **organ**.

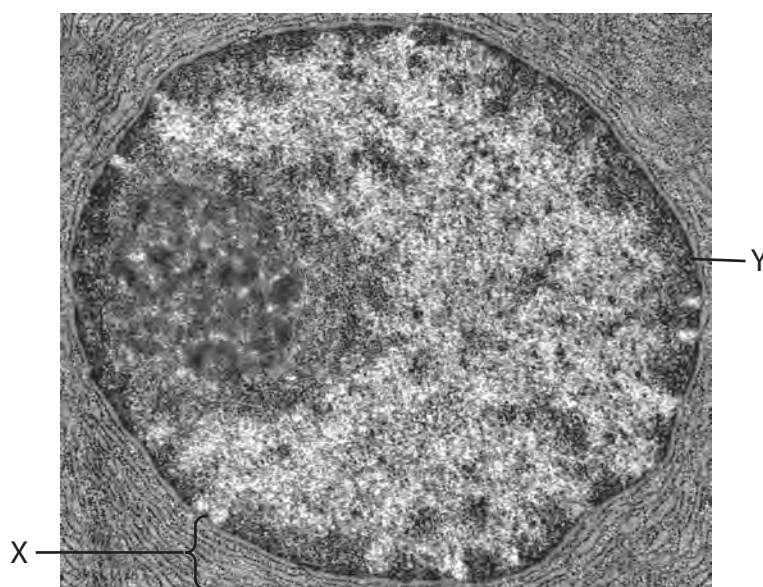
(1)

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(b) The photograph shows part of a cell.



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(i) Which organelle is labelled X?

(1)

- ☐ **A** chloroplast
- ☐ **B** endoplasmic reticulum
- ☐ **C** Golgi apparatus
- ☐ **D** plasmodesmata



(ii) The organelle surrounded by the membrane labelled Y contains the nucleolus.

Give the name and function of the organelle that contains the nucleolus.

(1)

Name

Function

(iii) Which of the following are functions of the nucleolus?

1. production of ribosomes
2. synthesis of DNA
3. synthesis of RNA

(1)

- ☐ **A** 1 and 2 only
- ☐ **B** 1 and 3 only
- ☐ **C** 2 and 3 only
- ☐ **D** 1, 2 and 3

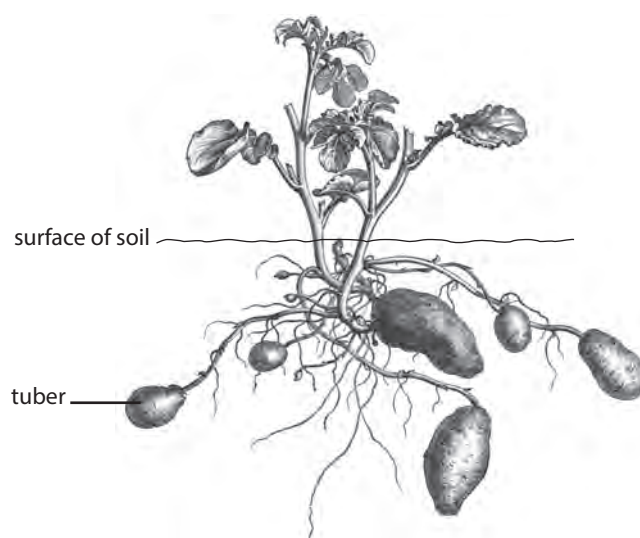
(iv) Organisms can be classified into three domains.

State which of the three domains would contain an organism with the organelles X and Y.

(1)



- (c) The drawing shows tubers of a potato plant developing below the surface of the soil.



Organic molecules are transported through the potato plant to the developing tubers.

These molecules can be used to form starch molecules.

- (i) Which structure in the cells of these tubers would store starch molecules?

(1)

- ☐ A amyloplast
- ☐ B chloroplast
- ☐ C middle lamella
- ☐ D vacuole

- (ii) Explain how the **structure** of starch enables the potato tubers to have a high energy content.

(2)

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(Total for Question 1 = 8 marks)

